

KitchenStock — Data Model

MongoDB / Mongoose · 25+ collections · every document scoped by **organizationId** (multi-tenant isolation)

FK = reference · [] = array · (poly) = polymorphic

Identity & Tenancy

- User**
owner / employee · organizationId (self) · permissions (RBAC) · tokenVersion
- Subscription**
owner → User · plan · status · aiOptIn · feature access
- Location**
organizationId · isActive
- NotificationPreference · PushToken**
per user · topic toggles · Expo tokens
- EmailSettings · OtpCode**
per org alerts · hashed OTP (TTL)
- CrmAuditLog · AccessOverride**
admin actions · per-org feature overrides

Catalog — Items & Recipes

- Item central**
FK Location · FK Supplier (supplierId) · stock · reorderPoint · units
- PackingMaterial**
FK Location · same unit model as Item
- Recipe central**
ingredients[] → Item · packingMaterials[] → PackingMaterial
subRecipes[] → Recipe (self) · costPerUnit · FK Location
- Menu**
recipes[] → Recipe · FK Location

Purchasing & Receiving

- Supplier**
catalog[] → Item (price quotes) · terms · lead time
- PurchaseOrder**
FK Supplier · lines[] → Item · status flow · FK Location
- Invoice**
FK Supplier · lines[] → Item · matchedPO → PurchaseOrder
source: manual | ai (OCR) · approve → stock + price
- InventoryReceiving**
items[] → Item · approve → stock

Movement & Sales

- SalesLog**
items[] → Recipe · deducts stock via recipe expansion · FK Location
- PosItemMap**
posName → Recipe · powers POS CSV import
- StockTransfer**
from / to Location · lines[] → Item / toItem
- ItemBatch**
FK Item · lot · expirationDate · remaining · discard → WasteLog
- SaleApproval**
pending staff sales · approve → deduct

Ledger & Audit

- StockHistory**
entity → Item / PackingMaterial (poly) · change · reason
- PriceHistory**
entity → Item / Packing / Recipe (poly) · price · label
- WasteLog**
FK Item · category · cost · deducts stock
- InventoryAdjustment**
entity → Item / Packing (poly) · delta · reason · audit
- + accountability feed**
aggregates these by user (who did what)

Modeling Conventions

- **Multi-tenancy**
every doc has organizationId; all queries scoped to it
- **Polymorphic ledgers**
{ entity, entityType } refs across Item / Packing / Recipe
- **Bill-of-materials**
Recipe expands recursively (sub-recipes) to base Items
- **Indexes & integrity**
compound org-scoped; unique per-org (supplier, poNumber, posName)
- **Soft-delete & audit**
isActive flags; stock & price changes are append-only history
- Two-tier units: purchase → tracking (conversion factors)

KEY FLOW: Supplier → PurchaseOrder → Receiving / Invoice → Item stock → Recipe → Sale → stock deduction → StockHistory + reports + forecast → auto-draft PO